Sri SAI RAM ENGINEERING COLLEGE
INSTITUTE OF TECHNOLOGY
West Tambaram, Chennai - 44

SAIRAM DIGITAL RESOURCES

YEAR III SEM V

24CSPW501

ARTIFICIAL INTELLIGENCE

UNIT NO 1

INTRODUCTION

1.4 Typical Intelligent Agents

- Agent
- Agent Program
- Environment
- Agent Types

COMPUTER SCIENCE & ENGINEERING

Outline

- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

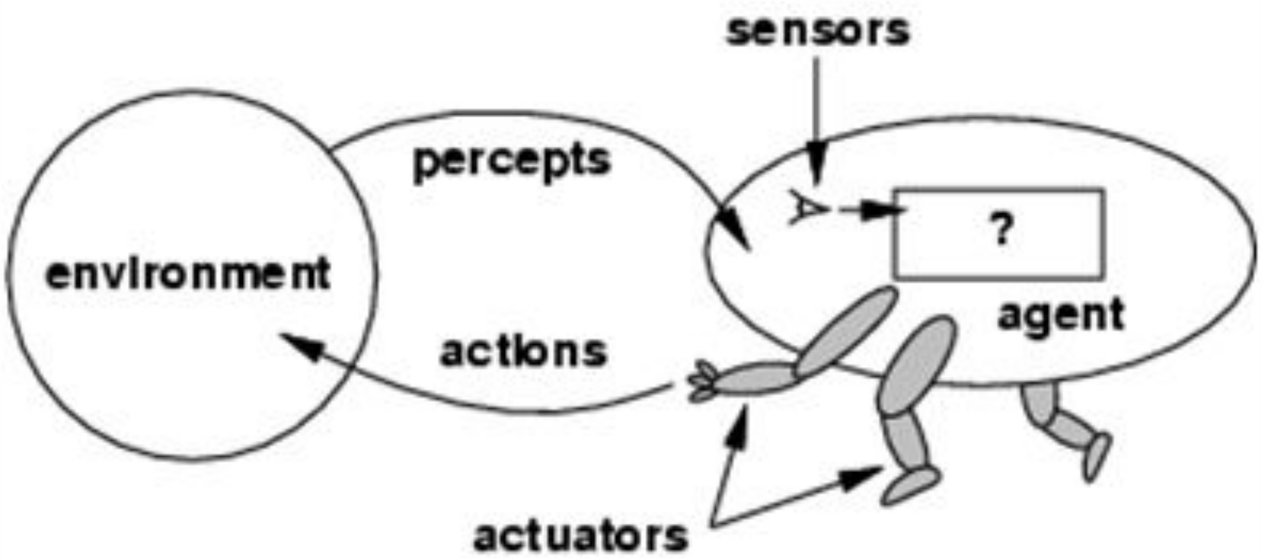
AGENT

•An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

•Human agent: eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators

•Robotic agent: cameras and infrared range finders for sensors; various motors for actuators

Agents and environments



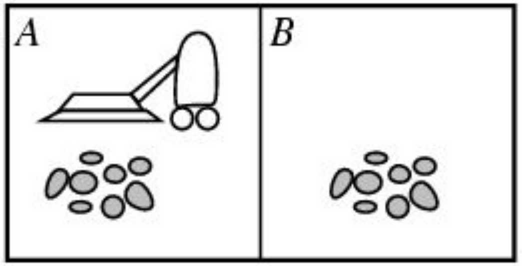
Agents and environments

•The agent function maps from percept histories to actions:

[*f*: P* à A]

•The agent program runs on the physical architecture to produce *f*
agent = architecture + program

Vacuum-cleaner world



•Percepts: location and contents, e.g., [A,Dirty]
Actions: *Left*, *Right*, *Suck*, *NoOp*

Rational agents

- An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful

- Performance measure: An objective criterion for success of an agent's behavior

E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rational agents

Rational Agent: For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Rational agents

•Rationality is distinct from omniscience (all-knowing with infinite knowledge)

•Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)

An agent is autonomous if its behavior is determined by its own experience (with ability to learn and adapt)

Agent functions and programs

- An agent is completely specified by the agent function mapping percept sequences to actions
- One agent function (or a small equivalence class) is rational

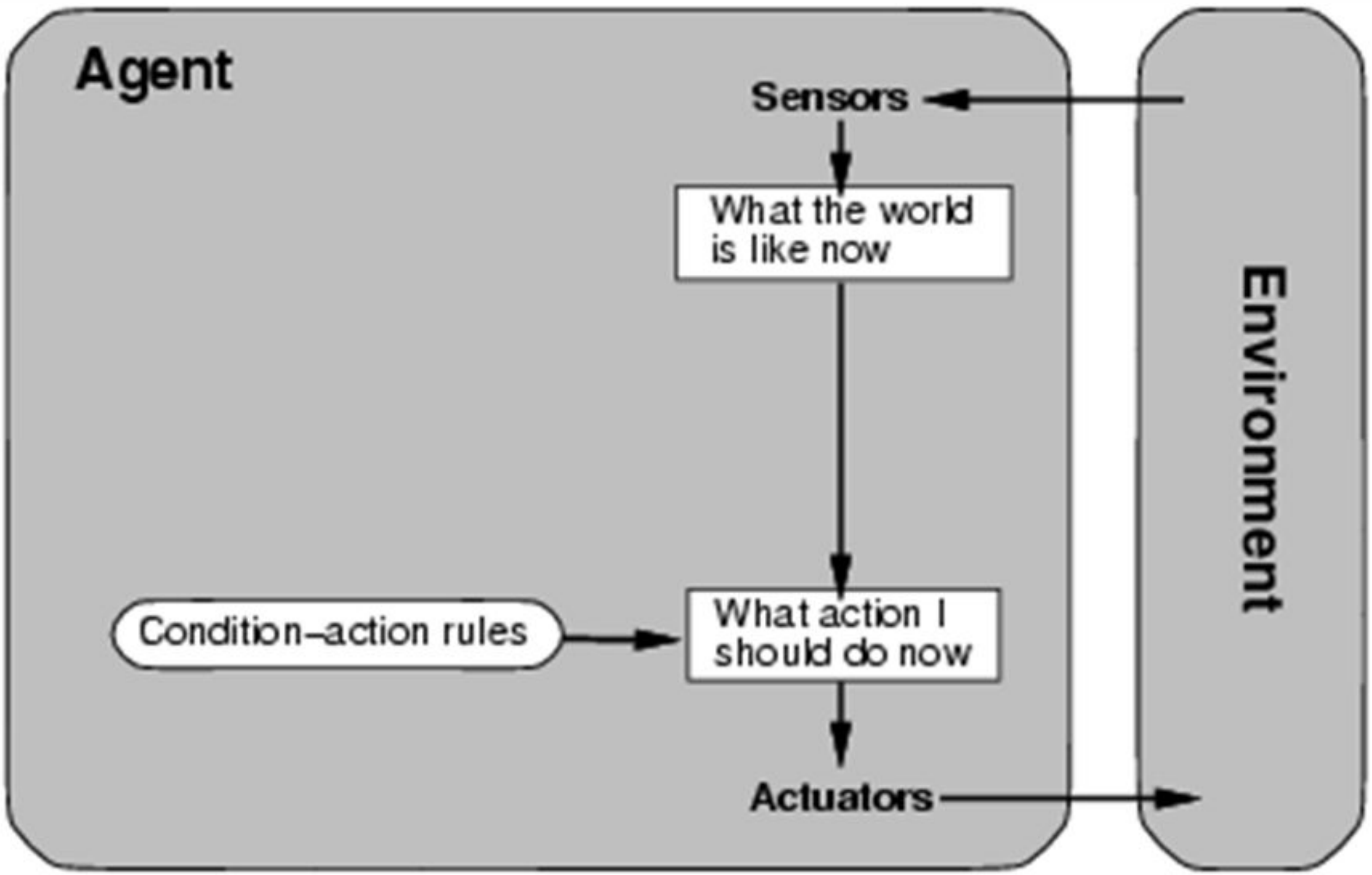
Aim: find a way to implement the rational agent function concisely

Agent types

Four basic types in order of increasing generality:

- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

Simple reflex agents



Simple reflex agents

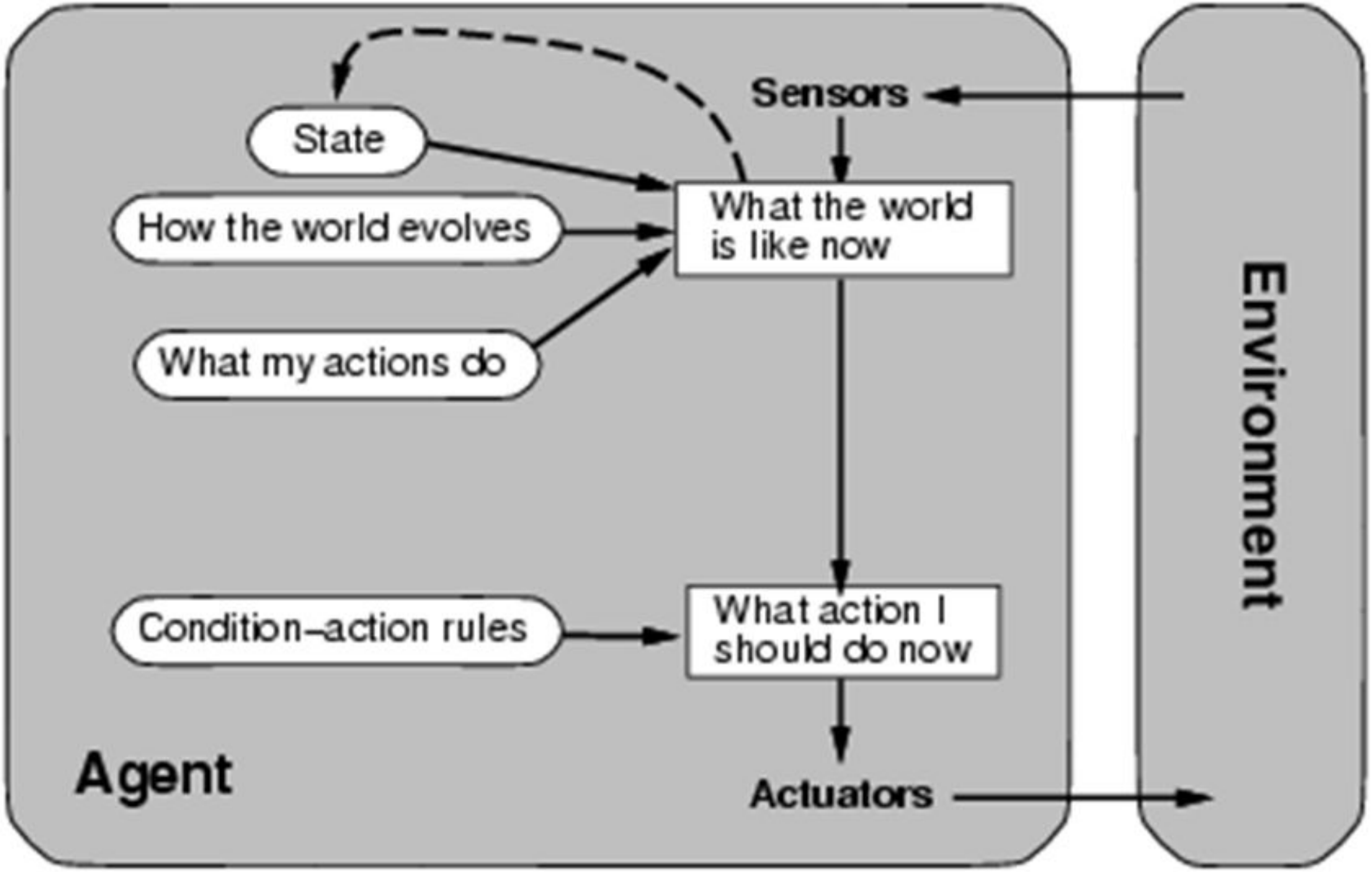
- Simple reflex agents act only on the basis of the current percept, ignoring the rest of the percept history.

- The agent function is based on the condition-action rule:

  if condition then action.
- Succeeds when the environment is fully observable.

- Some reflex agents can also contain information on their current state which allows them to disregard conditions.
- **Example**: A basic thermostat that turns on/off based only on current temperature.

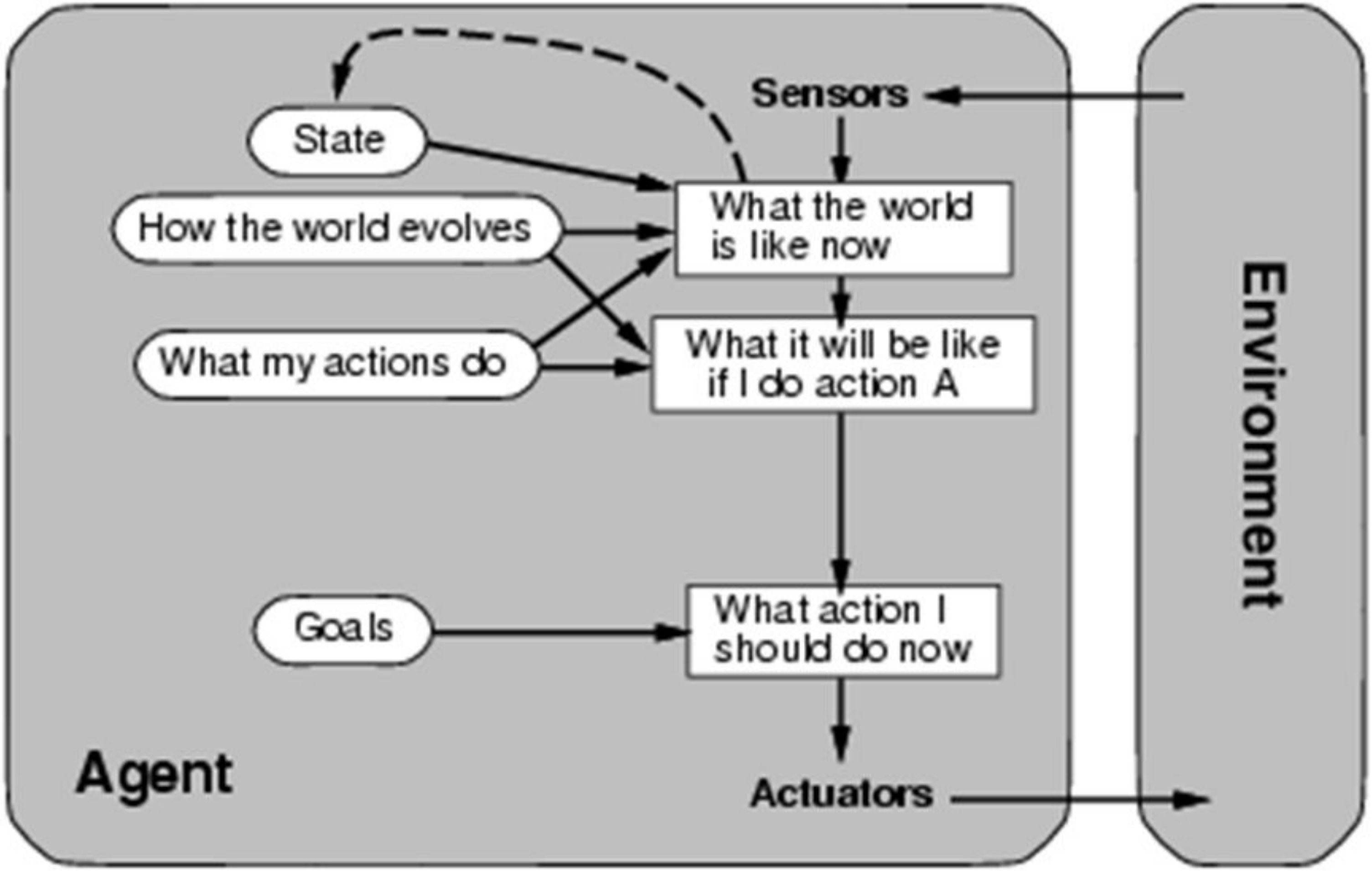
Model-based reflex agents



Model-based reflex agents

- A model-based agent can handle a partially observable environment.
- Its current state is stored inside the agent maintaining some kind of structure which describes the part of the world which cannot be seen.
- This knowledge about "how the world evolves" is called a model of the world, hence the name "model-based agent".
- **Example**: A robot vacuum cleaner mapping a room to know where it has cleaned.

Goal-based agents



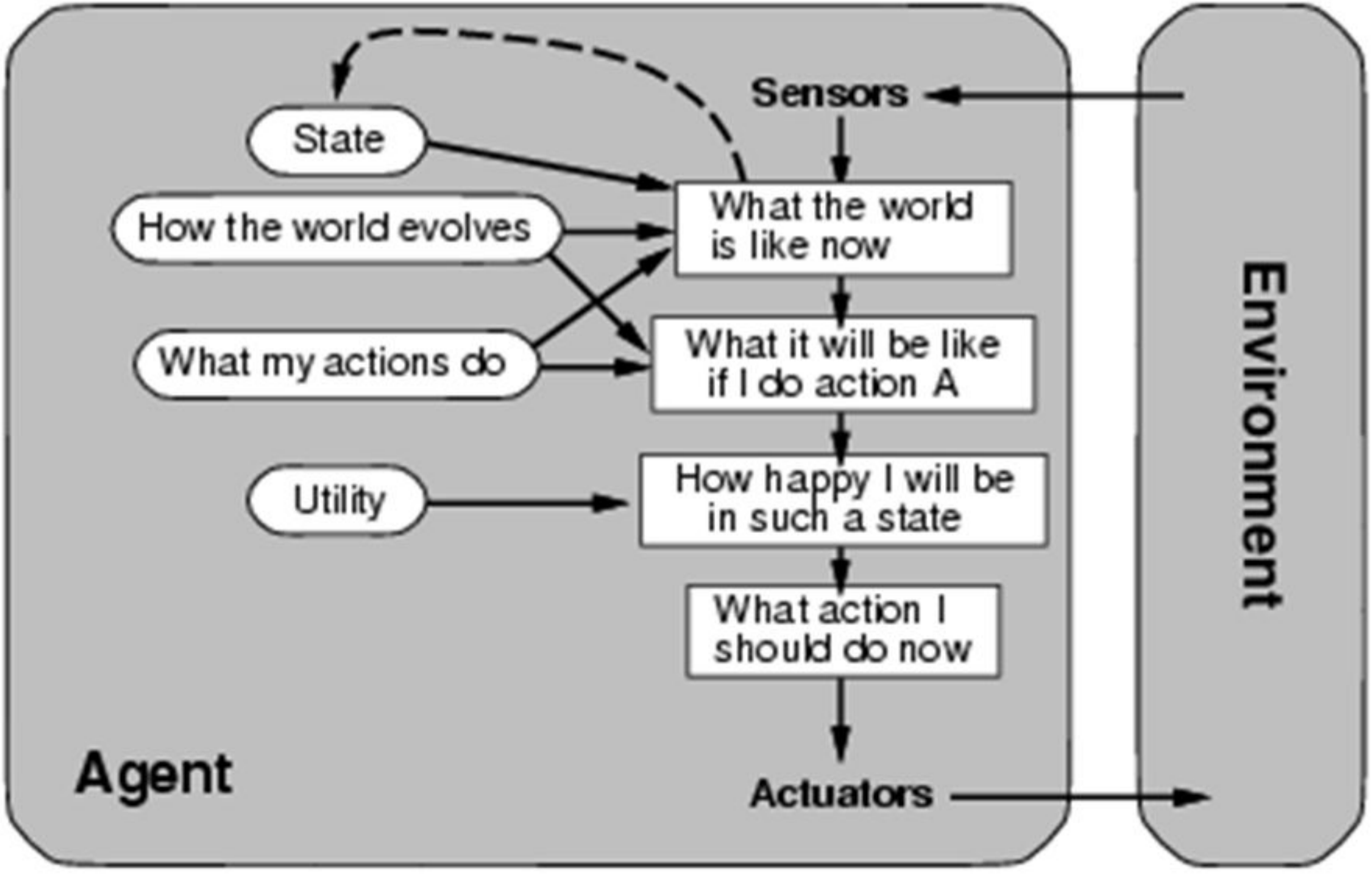
Goal-based agents

- Goal-based agents further expand on the capabilities of the model-based agents, by using "goal" information.

- Goal information describes situations that are desirable. This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state.

- Search and planning are the subfields of artificial intelligence devoted to finding action sequences that achieve the agent's goals.
- **Example**: A GPS navigation system finding the best route to a destination.

Utility-based agents



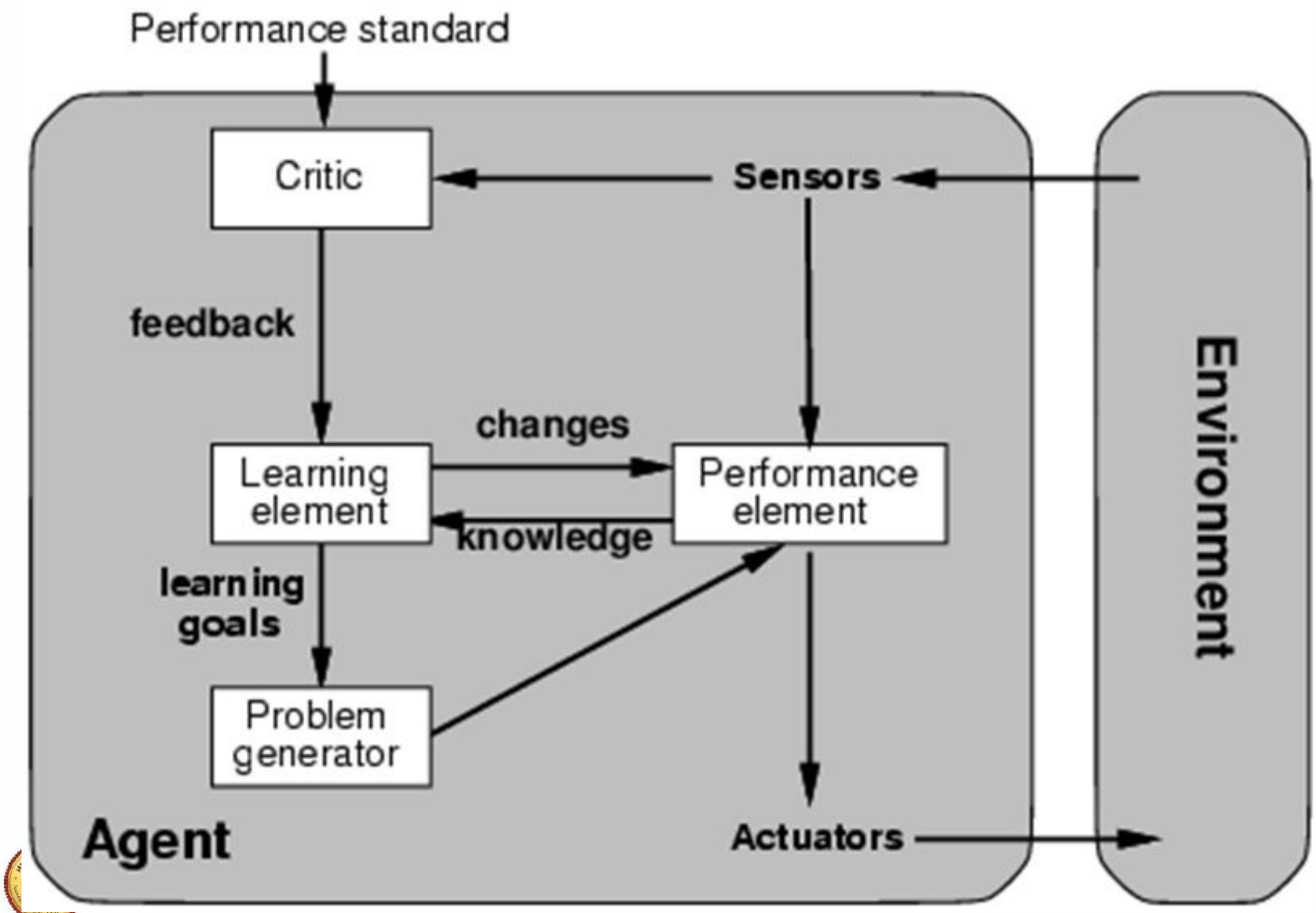
Utility-based agents

- Goal-based agents only distinguish between goal states and non-goal states.

- It is possible to define a measure of how desirable a particular state is. This measure can be obtained through the use of a utility function which maps a state to a measure of the utility of the state.

- A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent. The term utility, can be used to describe how "happy" the agent is.
- **Example**: A stock trading bot deciding trades to maximize profit, not just speed.

Learning agents



Learning agents

- Learning has an advantage that it allows the agents to initially operate in unknown environments and to become more competent than its initial knowledge alone might allow.
- The most important distinction is between the "learning element", which is responsible for making improvements, and the "performance element", which is responsible for selecting external actions.
- The learning element uses feedback from the "critic" on how the agent is doing and determines how the performance element should be modified to do better in the future.
- **Example**: An email spam filter that gets better at catching junk mail.

VIDEO LINKS

- https://www.youtube.com/watch?v=XO6SV0Mup1E
- https://www.youtube.com/watch?v=rWh9cK0ycuw